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SMART X-RAY LUGGAGE THREAT DETECTION SYSTEM USING YOLOV8 AND STM32

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Abstract : This paper presents a smart X-ray luggage threat detection system that combines artificial intelligence with embedded automation for real-time threat handling. A YOLOv8-based model is used to detect prohibited items such as guns, knives, and tools from live video streams. To improve reliability, a multi-frame confirmation approach is applied to reduce false alarms. Once a threat is confirmed, an STM32 microcontroller automatically controls the conveyor system, triggers alerts, and activates a rejection mechanism. The system also supports cloud-based monitoring for remote access. Experimental results show that the system achieves high detection accuracy while maintaining real-time performance, making it a practical and low-cost solution for automated baggage security.

IndexTerms - YOLOv8, X-ray Security System, Object Detection, STM32, Embedded Systems, Smart Surveillance, Conveyor Automation, Airport Security, Computer Vision, Real-Time Detection.

I. INTRODUCTION

The rapid growth of air travel and public transportation has significantly increased the demand for efficient baggage screening systems to ensure passenger safety. Airports and high-security facilities rely heavily on X-ray luggage inspection systems to detect prohibited items such as firearms, sharp objects, and hazardous tools. Conventional screening systems primarily depend on human operators to visually inspect X-ray images, which introduces limitations including operator fatigue, inconsistent judgment, delayed reaction time, and potential security risks.

Recent advancements in artificial intelligence and computer vision have enabled automated object detection systems capable of identifying threats with high accuracy. Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated remarkable performance in visual recognition tasks. Among these, the YOLO (You Only Look Once) family of object detection algorithms provides real-time detection capability while maintaining high precision, making it suitable for safety-critical applications.

Despite progress in AI-based detection, many existing research solutions remain limited to software-level detection without integration into physical response systems. Detecting a threat alone is insufficient unless immediate mechanical action is performed to isolate or remove the dangerous object. Therefore, combining intelligent vision systems with embedded automation is essential for building practical real-world security solutions.

To address these limitations, this work presents a smart X-ray luggage threat detection system that integrates AI-based detection with automated hardware response. A YOLOv8 model is used to analyze live video streams, while an STM32 microcontroller handles real-time control of the conveyor system and alert mechanisms. Unlike conventional approaches that only detect threats, the proposed system actively responds by isolating suspicious objects. A multi-frame validation strategy is also used to improve detection reliability and reduce false positives.

II. RELATED WORK

Automated threat detection in baggage screening systems has gained significant research attention due to increasing global security requirements. Traditional airport X-ray inspection systems primarily rely on human operators to analyze scanned images, which can result in reduced accuracy caused by fatigue, distraction, or subjective interpretation. To overcome these limitations, researchers have explored artificial intelligence–based approaches for automated prohibited object detection.

Early studies focused on classical image processing techniques such as edge detection, feature extraction, and template matching for identifying suspicious objects in X-ray images [11]. However, these methods suffered from poor robustness when objects overlapped or appeared under varying orientations and lighting conditions.

With the advancement of deep learning, convolutional neural networks (CNNs) became widely adopted for object recognition tasks [3]. Region-based detectors such as R-CNN, Fast R-CNN, and Faster R-CNN improved detection accuracy but required high computational resources, making real-time deployment challenging. Later, single-stage detectors like SSD and the YOLO family enabled faster inference speeds suitable for real-time applications [1].

Akcay et al. demonstrated that transfer learning using CNNs can effectively classify prohibited items within X-ray baggage security imagery, establishing a strong foundation for deep learning in this domain [13][14]. Further, Miao et al. introduced SIXray, a large-scale X-ray security benchmark specifically designed for prohibited item detection in overlapping and cluttered baggage images, highlighting the challenges of real-world deployment [14].

Recent research demonstrates the effectiveness of YOLO-based models in security surveillance and threat identification due to their balance between speed and accuracy. YOLOv3 introduced multi-scale detection improvements [17], while YOLOv7 achieved state-of-the-art real-time detection performance [16]. The latest YOLOv8 architecture further enhances performance through optimized feature extraction and efficient inference pipelines, enabling deployment in real-time safety environments [2].

Several works have also investigated intelligent surveillance systems integrated with IoT and cloud platforms for remote monitoring and data analytics [19]. While these approaches successfully detect threats, most implementations remain limited to software-level alerts without automatic mechanical response mechanisms.

Embedded system integration using microcontrollers such as Arduino or Raspberry Pi has been explored for automation tasks; however, many systems lack deterministic real-time control suitable for industrial environments [5]. Microcontroller platforms such as STM32 provide higher reliability, precise timing control, and efficient hardware interfacing for safety-critical automation.

Based on the literature review, several research gaps are identified. Many existing systems perform detection only without automated physical threat handling [14][15]. Limited work combines artificial intelligence detection, embedded control, and cloud monitoring into a unified architecture. Additionally, few implementations demonstrate a realistic conveyor-based X-ray simulation environment. Furthermore, false-positive reduction techniques in current systems are often insufficient for continuous real-time deployment [18].

To address these limitations, this work combines AI-based detection, embedded control, and cloud monitoring into a unified system. In addition to detecting threats, it performs real-time physical response, making it more suitable for practical deployment compared to existing approaches.

III. SYSTEM ARCHITECTURE

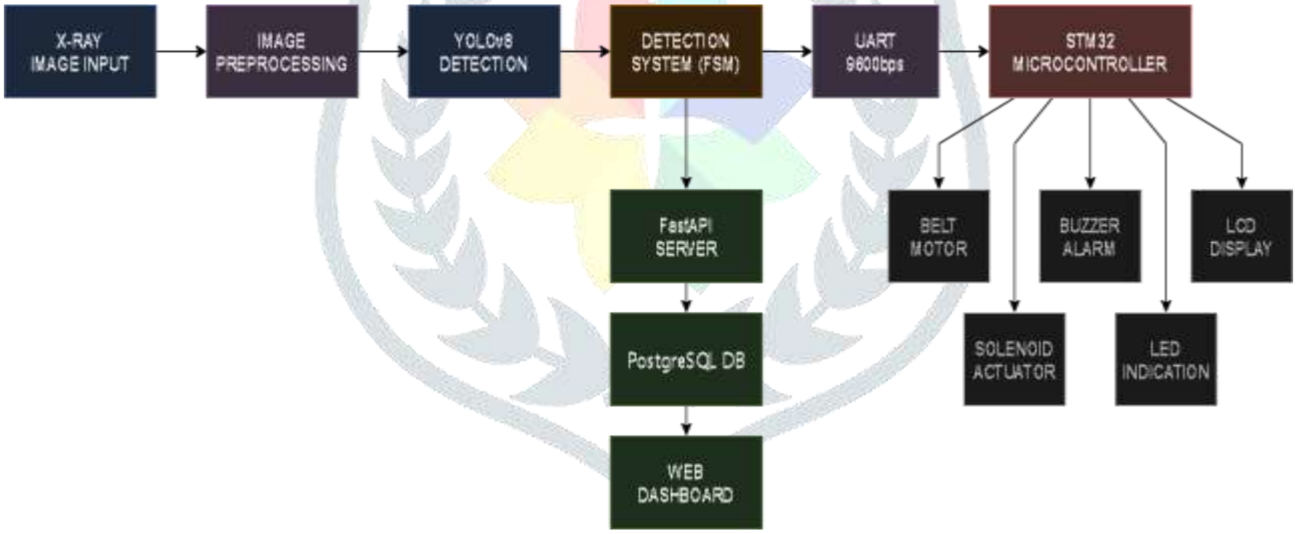

fig. 1 block diagram of the proposed smart x-ray luggage threat detection system integrating yolov8-based detection, stm32 embedded control, and cloud monitoring.

The proposed system combines artificial intelligence, embedded control, mechanical automation, and cloud connectivity into a unified real-time security platform. It is designed as a modular system consisting of image acquisition, detection, decision-making, hardware control, and cloud monitoring components. This structure allows efficient data flow and enables seamless integration between software and hardware modules.

a) Overall System Overview

A physical prototype replicating an airport X-ray baggage inspection system was developed using a conveyor belt mechanism enclosed within a custom-built structure made from wood, sticks, and cardboard to simulate a realistic scanning environment. A USB camera mounted above the conveyor continuously captures luggage images while objects move along the belt.

The captured video stream is processed on a GPU-enabled computer running the YOLOv8-based detection system. Based on the detection outcome, control signals are transmitted to the embedded hardware for automated response actions.

b) Image Acquisition and Pre-Processing

A USB webcam captures real-time video frames of luggage moving along the conveyor belt. Each frame is preprocessed to enhance image quality before being passed to the detection engine.

c) AI Detection Engine

The detection system is powered by a YOLOv8m model trained using the Ultralytics framework. The model was trained on a dataset of X-ray images containing five classes of prohibited items: gun, knife, scissors, pliers, and wrench. During operation, the model processes video frames in real time and generates bounding boxes along with confidence scores. To improve reliability, a multi-frame confirmation strategy is used, where detections must remain consistent across multiple frames before being classified as a threat.

d) Decision and Communication Module

Once a threat is confirmed, the system generates a decision signal where “0” indicates safe operation and “1” indicates a detected threat.

e) Embedded Hardware Controller

The STM32F103C8T6 (Bluepill) microcontroller serves as the embedded controller, managing all output actuators and indicators based on serial commands received from the detection system. It operates in three modes: Safe, Danger, and Emergency Stop.

Operating Modes:

The system operates in three modes. In Safe Mode, the conveyor belt runs continuously, the solenoid remains off, the green LED is active, the buzzer remains off, and the LCD displays a safe status. In Danger Mode, the conveyor belt stops, the solenoid activates to eject the object, the buzzer alarm is triggered, the red LED turns on, and the LCD displays a danger alert. In Emergency Stop Mode, triggered by a hardware push button, the system immediately shuts down all motion, activates the alarm, and requires manual reset.

f) Mechanical Automation Subsystem

A 12V DC motor drives a conveyor belt of approximately 4×26 inches. Motor speed is regulated using a PWM controller module, while DPST switches allow direction control.

A 12V solenoid actuator, controlled through a relay, mechanically pushes detected threat objects away from the conveyor path, simulating real industrial sorting mechanisms used in security systems.

g) Power Management Architecture

The system uses a dual-rail power architecture, supplying 12V for high-power actuators and a regulated 5V for logic-level components, ensuring noise isolation between subsystems.

h) Cloud Monitoring and Data Synchronization

Detection events are transmitted to a cloud backend, enabling remote monitoring through a secured web dashboard with real-time status updates and historical log access.

IV. IMPLEMENTATION

The system was implemented by integrating AI-based detection, embedded firmware, mechanical components, and cloud monitoring into a single working platform. This setup enables real-time threat detection along with immediate hardware response.

a. AI Model Development and Training

The object detection model was developed using the Ultralytics YOLOv8 framework in a Python environment configured through Anaconda. Training was performed locally on a system equipped with an NVIDIA RTX2060 GPU, enabling accelerated deep learning computation.

A publicly available airport X-ray dataset containing prohibited items was used for training. The dataset consists of 8,295 labeled images, including 5,806 training images, 1,660 validation images, and 829 testing images. All images were resized to a resolution of 640 × 640 prior to training.

The YOLOv8m architecture was selected to balance detection accuracy and real-time performance. The model was trained for 35 epochs to detect five classes of prohibited objects, namely gun, knife, scissors, pliers, and wrench. During inference, the trained model generates bounding boxes along with confidence scores, and a threshold of 0.80 is applied to reduce incorrect detections.

b. Real-Time Detection Software

The detection software was developed using Python and OpenCV, where a USB webcam continuously streams video frames using the cv2.VideoCapture interface. The software processes video in multiple stages, including frame capture, image preprocessing (brightness adjustment, rotation, and sharpening), YOLOv8 inference, and decision generation. A multi-frame confirmation buffer is used to ensure stable and reliable detection. A 12-frame FSM-based confirmation algorithm ensures that only consistent detections trigger hardware actions, significantly reducing false positives. The application operates as a desktop system using Flask backend integration and a pywebview graphical interface for local monitoring.

c. Embedded Firmware Implementation

The embedded control system was programmed on an STM32F103C8T6 Bluepill microcontroller using UART communication for interaction with the AI detection system. The firmware continuously listens for serial commands, where “0” indicates SAFE MODE and “1” indicates DANGER MODE, and accordingly controls digital output pins connected to relays, indicators, and actuators.

Hardware Control Logic:

In Safe Mode, the conveyor motor relay remains enabled, the solenoid is disabled, the green LED is active, the buzzer remains off, and the LCD displays a safe system status. In Danger Mode, the conveyor motor stops, the solenoid actuator extends to eject the object, the buzzer activates at 1 kHz, the red LED turns on, and the LCD displays a danger alert. In Emergency Stop Mode, triggered by a physical push button, all motion systems shut down immediately and an alarm notification is activated.

d. Conveyor Belt and Mechanical System

A prototype conveyor belt system measuring approximately 4×26 inches was constructed to simulate real baggage movement. A 12V DC motor drives the belt, while speed control is achieved through an external PWM motor controller.

Motor direction can be manually changed using DPST switches. When a threat is detected, the STM32 disables motor motion and activates a 12V solenoid actuator, which physically pushes the detected object away from the conveyor path.

This mechanism demonstrates automated threat isolation similar to industrial sorting systems.

e. Power Distribution and Electronics Integration

The system operates using a 230V AC mains supply converted through an SMPS module into 12V DC. A buck converter further regulates this to 5V for logic components. The 5V supply powers the STM32, LCD, LEDs, buzzer, and relays, while the 12V supply powers the motor and solenoid actuators. This architecture ensures stable performance and minimizes electrical noise interference.

f. Cloud Integration and Monitoring Platform

To enable remote supervision, detection events are transmitted to a cloud server using WebSocket communication. The cloud system includes a FastAPI backend server, a PostgreSQL database, snapshot storage of detected threats, JWT-secured authentication, and a web-based dashboard. The dashboard supports live detection monitoring, event history logging, snapshot visualization, and remote access through web and mobile platforms.

V. RESULTS & DISCUSSION

The performance of the proposed Smart X-Ray Luggage Threat Detection System was evaluated based on detection accuracy, real-time responsiveness, and automated hardware response reliability. Both the artificial intelligence model and embedded automation system were experimentally validated using the developed prototype environment.

a. Model Training Performance

The YOLOv8m model was trained using 8,295 annotated X-ray images for 35 epochs on an NVIDIA RTX2060 GPU. The model achieved an mAP@50 of 0.91, an inference speed of ~25 FPS, and operated with a confidence threshold of 0.80, demonstrating strong detection performance suitable for real-time applications.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{F1-score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

$$\text{mAP} = \frac{1}{N} \times \sum \text{AP}_i$$

table 1: performance metrics of detection model

Class	Precision	Recall	F1-Score	mAP@50
Gun	0.94	0.87	0.90	0.92
Knife	0.93	0.85	0.89	0.91
Pliers	0.92	0.86	0.89	0.90
Scissors	0.91	0.84	0.87	0.89
Wrench	0.92	0.87	0.89	0.91
Overall (Mean)	**0.924**	**0.858**	**0.888**	**0.91**

The results show that the system achieves high detection accuracy across all object categories, with mAP@50 values above 0.89. Objects such as guns and knives are detected with higher confidence due to their distinct shapes, while minor confusion occurs between similar tools like pliers and wrenches. The use of multi-frame validation significantly reduces false detections, improving overall system reliability.

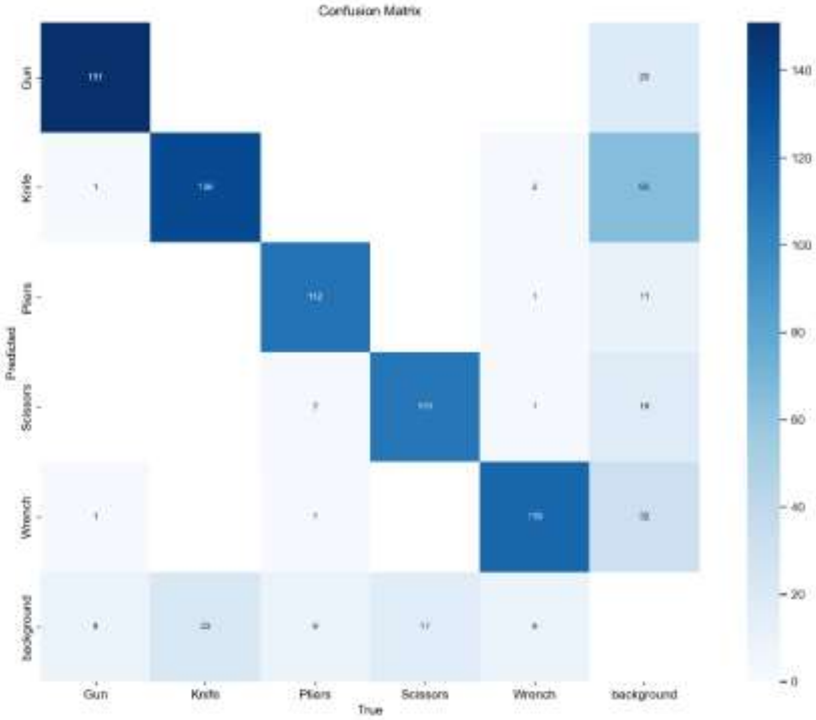

fig. 2 confusion matrix representing classification accuracy of the yolov8 model.

b. Detection Accuracy Analysis

Performance evaluation using the testing dataset of 829 images showed high classification reliability. Guns and knives achieved the highest detection confidence due to distinct shapes, while slight overlap between tool-based objects such as pliers and wrenches caused minor misclassifications in rare cases. The multi-frame confirmation mechanism significantly reduced false positives by ensuring consistent detections across frames.

c. Real-Time System Performance

The system operated reliably under real-time conditions with continuous conveyor movement. It maintained stable detection at ~25 FPS, achieved near-instant UART communication, ensured reliable switching between SAFE and DANGER modes, and maintained consistent synchronization between AI decisions and embedded hardware response.

d. Hardware Response Validation

The embedded automation system demonstrated reliable performance across all operating modes. In Safe Mode, the conveyor operated continuously with green LED indication and no alarm. In Danger Mode, the conveyor stopped, the solenoid ejected the object, the buzzer activated, and the LCD displayed a warning. In Emergency Stop Mode, the system successfully halted all motion and entered a safe shutdown state. Repeated testing confirmed consistent and stable operation.

fig. 3 hardware implementation of the proposed system.

e. System Latency Analysis

The end-to-end detection-to-response latency of the system was measured under real-time operating conditions. Latency is defined as the time interval between threat confirmation and the activation of the STM32-controlled response. The system achieves an overall latency of 150–200 ms, including detection, communication, and actuation delays. This response time is suitable for real-time conveyor-based security applications.

f. Cloud Monitoring Evaluation

The cloud synchronization module successfully uploaded detection snapshots and logs to the FastAPI server. The dashboard enabled real-time monitoring, historical data access, and remote system interaction, demonstrating scalability for centralized security applications.

fig. 4 system dashboard showing safe state detection.

fig. 5 system dashboard showing danger state detection.

VI. CONCLUSION

This study demonstrates that integrating deep learning-based object detection with embedded hardware automation is a viable approach for real-time baggage security. The proposed system successfully identified prohibited items across five threat categories such as guns, knives, scissors, pliers, and wrenches — achieving a mAP@50 of 0.91 at ~25 FPS.

A key finding is that software-only detection is insufficient for practical deployment. The addition of STM32-based embedded control, conveyor automation, and solenoid-based rejection transforms detection into a fully autonomous response platform. The multi-frame FSM confirmation strategy effectively suppressed false positives, and an end-to-end latency of 150–200 ms confirms suitability for real-world conveyor environments. Cloud monitoring further enables centralized supervision across multiple checkpoints.

Overall, this system provides a scalable solution that bridges the gap between software analysis and physical threat handling. Future work will focus on multi-camera integration, improved model generalization, and edge AI deployment.

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